Inventory Management

The full-domain reference · mrsupplychain.ai

The working capital dial — balancing service, cost, and cash across the network.

Inventory Management is the discipline that decides how much capital sits idle in the form of stock, where that stock should live in the network, and under what rules it should move. Every excess unit is a tax on the balance sheet; every shortage is a tax on service and revenue.

HOW TO READ THIS DOMAIN

1 · Inventory Policy & Segmentation. The rules that decide how much of each SKU the network is allowed to hold

2 · Multi-Echelon & Network Inventory. Where inventory actually lives and how it moves across the network

3 · Demand-Driven Replenishment. Pulling inventory based on actual consumption rather than forecast-driven push

4 · MRO & Critical Spares Inventory. Protecting uptime with a different economic logic than finished goods

5 · Inventory Analytics & Technology. Turning inventory data into decisions and financial language

STAGE 1 · POLICY · 01 OF 06

Inventory Policy & Segmentation

This is the foundational layer. Before you can optimize positions or automate replenishment, you must decide — deliberately and differently for different items — how much inventory the system is allowed to hold.

01 ABC-XYZ segmentation & differentiated inventory policies

ABC-XYZ segmentation classifies every SKU on two independent dimensions — value contribution (ABC) and demand variability (XYZ) — producing a nine-cell matrix where each cell gets its own inventory policy: service target, planning method, review frequency and ownership.

Applying one policy to every SKU over-protects stable, low-value items and under-protects volatile, high-value ones. **Differentiation is the core principle of inventory management: the right policy depends on the item's value and predictability (Silver, Pyke and Thomas — Inventory and Production Management in Supply Chains, CRC Press)**. The matrix turns that principle into explicit rules planners and systems can follow.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A discrete manufacturer with 4,800 active SKUs found that the AZ cell (187 items) accounted for 41% of inventory value and 63% of stockout incidents. A differentiated policy reduced total inventory 18% while improving OTIF on A items by 4.2 points.

WHAT YOU WOULD USE

> **Differentiated policy is not a refinement; it is the policy.** Classify by value and variability, assign rules per cell, load them into the system and reclassify every quarter.

02 Safety stock calculation & service level optimization

Safety stock is the buffer held against uncertainty in demand and supply. A defensible calculation links a chosen service target, measured demand variability and lead-time variability into one number per item and location.

Safety stock is usually the largest controllable part of inventory investment, and its cost rises non-linearly with the service target: **each additional point of service near the top of the curve costs more inventory than the last**. Choosing the metric matters too — cycle service level and fill rate give very different numbers for the same item **(Silver, Pyke and Thomas — Inventory and Production Management in Supply Chains)**.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> An industrial equipment manufacturer applied a uniform 95% service level. After segmenting and resetting targets, total safety stock value fell 22% while fill rate on A items rose from 93.1% to 97.4%.

WHAT YOU WOULD USE

> **Price the service level before you promise it.** Choose the metric, use forecast error and lead-time variability, differentiate by segment and recalculate every cycle.

03 Reorder point, EOQ & replenishment parameters

The reorder point (ROP) answers when to order; the economic order quantity (EOQ) answers how much. Together with review frequency and real lead times, they are the operational engine of replenishment.

The formulas are a century old — EOQ dates to Harris (1913) — and still sound; **what fails in practice is the data feeding them**. Lead times and order quantities that were true years ago produce excess and shortage at the same time, and nobody notices because the parameters look official in the ERP.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A process manufacturer found that 38% of its reorder points were based on lead times from six years earlier. After a full parameter cleanse, average inventory fell 14% and expedite orders dropped 31%.

WHAT YOU WOULD USE

> **Good formulas, stale data, bad results.** Audit lead times, refresh EOQ and ROP with current inputs, and flag drift automatically.

04 Inventory reduction programs & working capital liberation

An inventory reduction program is a structured, time-bound initiative that removes excess inventory through a deliberate sequence of levers while protecting service with explicit guardrails.

Excess inventory ties up cash and hides problems, but cuts made without understanding why the stock exists quickly turn into shortages and are reversed. **The sequence matters: obsolete stock and over-sized buffers can be removed safely; cutting the wrong items damages service.** Working capital released this way is durable because the causes are fixed, not just the balances.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A consumer durables company reduced inventory by $46.2M while finishing at 97.1% OTIF on A items by sequencing the levers. A previous flat percentage cut had been largely reversed within nine months.

WHAT YOU WOULD USE

> **Cut causes, not percentages.** Diagnose the stock, protect service with hard guardrails, sequence the levers and track both sides every week.

05 Slow-moving & obsolete (SLOB) inventory management

SLOB management is the disciplined process of identifying, classifying, disposing of and — most importantly — preventing slow-moving and obsolete inventory.

SLOB is the visible symptom of upstream failures: optimistic forecasts, uncontrolled phase-outs, engineering changes without inventory review, and minimum order quantities that exceed demand. **Every new unit of SLOB is evidence that the process creating it is still running.** Disposal recovers some value; prevention is where the money is.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A medical device company reduced new SLOB creation by 74% after creating a mandatory phase-out checklist tied to engineering change orders.

WHAT YOU WOULD USE

Dispose of the past, prevent the future. Classify by age, recover value through a disposition hierarchy, and stop new SLOB at its source in planning and engineering.

Multi-Echelon & Network Inventory

Once the policy rules exist, the next question is physical: where should the stock sit? Multi-echelon thinking converts inventory from a collection of local buffers into a coordinated network asset.

01 Multi-echelon inventory optimization across the network

Multi-echelon inventory optimization (MEIO) calculates inventory positions for all stocking points of a network at once — plants, central and regional DCs — instead of optimizing each node in isolation.

When each node protects itself, the network holds safety stock several times over against the same uncertainty. **Optimizing jointly lets buffers sit where they protect the most service per unit of inventory**, often upstream where demand is pooled **(Graves and Willems, 2000 — Manufacturing and Service Operations Management)**. The gain depends on network structure and data quality, so it should be modeled, not assumed.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A consumer electronics company with one central DC and six regional DCs reduced total network inventory 27% while improving fill rate 1.8 points after implementing multi-echelon optimization.

WHAT YOU WOULD USE

> **Optimize the network, not the nodes.** Map the topology, pool risk where it pays, set service at the customer-facing edge and let the model place the buffers.

02 Distribution center inventory management & replenishment

DC inventory management defines how stock is positioned and replenished in distribution centers — push, pull or hybrid models, review cycles and cross-docking rules — by product segment.

Distribution centers often hold the largest concentration of finished-goods inventory. **Push models position stock on forecast and work for long lead times and stable demand; pull models replenish on consumption and work for fast, variable items.** Using a single model for everything guarantees the wrong one for part of the portfolio.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A food distributor moved its top 18% of volume to a pull and cross-dock model and reduced total DC inventory 19% while raising regional service level 3.1 points.

WHAT YOU WOULD USE

> **Match the replenishment model to the item.** Segment at the DC, push where forecast works, pull where consumption is fast, and flow what is synchronized.

03 Inventory in transit management & pipeline visibility

In-transit inventory is stock that has left the supplier or an upstream node but has not yet been received. Pipeline visibility makes it available to planning as confirmed supply with a reliable arrival date.

With long international lead times, a large share of inventory can be on the water or on the road at any moment. **What the planning system cannot see, it cannot plan: it either orders again or fails to protect against a late arrival.** Pipeline

inventory is also a strategic position that can be redirected when demand shifts.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A global industrial company reduced total inventory 9% and cut expedite freight 34% after implementing systematic ASN capture and including the pipeline in its APS.

WHAT YOU WOULD USE

> **Make the pipeline visible and plannable.** Capture ASNs, track milestones, net confirmed in-transit supply in planning and manage it as a position you can redirect.

04 Seasonal & promotional inventory planning

Seasonal and promotional inventory planning builds stock ahead of known demand peaks and manages the drawdown afterward, so residual inventory does not become slow-moving or obsolete.

Pre-building is necessary when capacity or lead time cannot follow the peak, but peak demand is the hardest to forecast. **The right build size balances the cost of running short against the cost of leftovers — the logic of the newsvendor model (Fisher, Hammond, Obermeyer and Raman, 1994 — Harvard Business Review)**. Without an exit plan, leftovers become the next SLOB problem.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A seasonal outdoor products company reduced residual inventory from 18–25% to 7% of the build and cut related write-offs 61% after instituting formal exit plans.

WHAT YOU WOULD USE

> **Build for a range, exit by design.** Size the pre-build on scenarios and costs, write the exit plan upfront and learn from every season.

05 Consignment inventory & vendor-managed inventory (VMI)

Consignment and vendor-managed inventory (VMI) are collaborative models in which the supplier owns or actively manages inventory at the customer's location, based on shared consumption data and agreed limits.

Designed well, these models reduce the customer's working capital and the supplier's demand uncertainty at the same time, because the supplier plans from real consumption instead of erratic orders **(Waller, Johnson and Davis, 1999 — Journal of Business Logistics)**. **Designed poorly, they just move the inventory from one balance sheet to the other.**

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A manufacturer moved its top 40 MRO and packaging items to VMI. Its inventory on those items fell 34% and stockouts dropped to near zero because both parties ran it as a joint process.

WHAT YOU WOULD USE

Share the data, not just the stock. Choose suitable categories, contract the rules, automate consumption data and govern the model together.

Demand-Driven Replenishment

Classic MRP amplifies variability. Demand-driven approaches decouple the levels, react to actual consumption, and convert inventory into a deliberate, flow-oriented asset.

01 Demand-Driven MRP (DDMRP) methodology

Demand Driven MRP (DDMRP) places strategic inventory buffers at decoupling points in the bill of materials and network, sizes them dynamically, and replenishes them from actual demand and buffer status rather than from forecast-driven MRP netting.

Classic MRP passes every forecast change through the whole bill of materials, which amplifies variability and creates nervous plans. **Decoupling buffers absorb that variability at chosen points, so lead times upstream and downstream of each buffer shrink (Ptak and Smith — Demand Driven Material Requirements Planning, Industrial Press)**. Results depend on buffer positioning and on planners adopting a different way of working.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A discrete manufacturer implemented DDMRP on its top 420 items. Inventory fell 31%, OTIF rose from 91% to 97.4%, and expedite orders dropped 64%.

WHAT YOU WOULD USE

> **Decouple, then pull.** Position buffers deliberately, size them dynamically, execute on net flow and buffer status, and invest in the change of planning practice.

02 Pull systems: kanban design & implementation

Kanban is a visual or electronic pull system that authorizes production or movement of material only when a downstream process signals consumption, limiting work in process by design.

Kanban comes from the Toyota Production System, where it was designed for stable, repetitive production **(Ohno — Toyota Production System: Beyond Large-Scale Production)**. **Under those conditions it cuts work in process and lead time with very little planning overhead.** Forced onto highly variable or low-volume demand, it creates shortages, oversized loops and workarounds.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> An automotive components plant implemented kanban on 85 high-volume, stable part numbers. Work in process fell 41% and average replenishment lead time dropped from 6.2 to 1.8 days.

WHAT YOU WOULD USE

> **Pull where demand is stable.** Choose X items, size the loops properly, separate production from transport signals and keep recalculating the cards.

03 Dynamic replenishment & automated reorder systems

Dynamic replenishment continuously recalculates reorder parameters from recent demand and lead-time data and generates replenishment orders automatically for eligible items, leaving planners to manage exceptions.

Planners spend much of their time on routine reorders that a system could handle. **But automation on static parameters only executes outdated decisions faster.** The value comes from combining automatic order generation with parameters that keep themselves current, and with clear rules for which items qualify.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A distribution company automated replenishment on 6,200 of 11,400 SKUs. Planner time on routine reordering fell about 60%, and inventory on automated items decreased 11% while service rose 1.4 points.

WHAT YOU WOULD USE

> **Automate the routine, keep humans on the exceptions.** Qualify items, refresh parameters every cycle, manage by exception and measure how the automation performs.

04 Omnichannel inventory & available-to-promise (ATP)

Omnichannel available-to-promise (ATP) is the real-time capability to commit a customer order against the true available inventory across all relevant nodes and channels, applying explicit allocation and priority rules.

In omnichannel, the same unit can be sold through a store, a website and a marketplace. **Without one accurate ATP, the company either oversells and cancels, or hides stock in channel silos and loses sales.** The allocation rules are a commercial decision; the engine only executes them.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A specialty retailer reduced oversell incidents 78% and improved overall availability after implementing a unified ATP engine with clear channel priority rules.

WHAT YOU WOULD USE

> **One inventory, one promise engine, explicit rules.** Unify availability, let the business set allocation priorities and keep the engine in sync with every order event.

05 Retail & point-of-sale inventory management

Retail and point-of-sale inventory management focuses on the last meter of availability: true on-shelf availability, detection of phantom inventory, and the use of sell-out data as the primary demand signal.

The shopper only experiences what is on the shelf. Studies of store records show that system inventory is frequently wrong, and phantom stock — units the system believes are there but are not — blocks replenishment exactly when it is needed **(DeHoratius and Raman, 2008 — Management Science)**. **Out-of-stocks then distort sell-out data and upstream forecasts.**

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A grocery retailer raised on-shelf availability from 91% to 96.5% and reduced estimated lost sales 28% after focusing on phantom inventory and redesigning store replenishment priorities.

WHAT YOU WOULD USE

Manage the shelf, not just the system. Measure on-shelf availability, hunt phantom stock, correct sell-out for stockouts and design replenishment to keep the shelf full.

STAGE 4 · CONTROL · 04 OF 06

MRO & Critical Spares Inventory

MRO follows different economics. The cost of not having a critical spare almost always dwarfs the cost of holding it — provided you hold the right parts.

01 Critical spare parts classification & stocking policy

Critical spare parts classification replaces value-based ABC with a framework driven by the consequence of failure, supply lead time and failure probability, and sets a stocking policy for each class.

Spare parts behave differently from production inventory: demand is sporadic, and the cost of not having a part is downtime, not a lost sale **(Huiskonen, 2001 — International Journal of Production Economics)**. **Value-based ABC under-protects cheap parts whose absence can stop an entire line.** A low-cost bearing on a bottleneck asset can matter far more than an expensive assembly on a redundant one.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A continuous-process plant identified 312 high-consequence, long-lead-time parts that were 4% of item count but linked to 61% of historical downtime. Raising their protection while reducing stock on low-consequence items cut MRO inventory 18% and related downtime events 57%.

WHAT YOU WOULD USE

> **Classify spares by consequence, not by price.** Score assets, map parts, stock by criticality and lead time, and revisit the matrix when the plant changes.

02 Reliability-centered maintenance (RCM) & inventory alignment

RCM–inventory alignment links the maintenance strategy chosen for each asset — preventive, predictive, run-to-failure — to the spare parts policy that supports it.

Reliability-centered maintenance decides, failure mode by failure mode, how each asset should be maintained **(Moubray — Reliability-Centered Maintenance)**. **If spare parts policy is set independently, plants end up holding parts for strategies that no longer exist and missing parts for tasks that are due.**

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A heavy-industry site found 22% of MRO inventory value tied to abandoned maintenance strategies. After alignment, MRO inventory fell 24% and planned jobs delayed by missing parts dropped from 11% to under 3%.

WHAT YOU WOULD USE

> **Let the maintenance strategy set the spares policy.** Start from RCM, link tasks to parts, stock by strategy and keep CMMS and inventory in sync.

03 Spare parts obsolescence & lifecycle management

Spare parts obsolescence management tracks the lifecycle of equipment and its parts so that end-of-support, last-time-buy and replacement decisions are made proactively, not in an emergency.

When a critical spare becomes unavailable, the plant faces redesign, reverse engineering or unplanned equipment replacement under time pressure. **Obsolescence is largely predictable from manufacturer notices and equipment age**, and there is an international standard for managing it **(IEC 62402 — Obsolescence management)**.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A process manufacturer received a last-time-buy notice with a 14-month window. Because it had a living map and a defined process, it evaluated options with full data and avoided an emergency redesign.

WHAT YOU WOULD USE

> **See end-of-life coming.** Keep the equipment-to-parts map alive, watch manufacturer notices, decide last-time buys with data and retire spares with their assets.

04 Condition-based inventory & predictive maintenance integration

Condition-based inventory uses predictive maintenance signals — vibration, temperature, oil analysis — to trigger purchasing or staging of spare parts when degradation is detected, instead of holding static safety stock.

Static stock for parts that fail rarely can sit for years. **When a failure mode gives a reliable warning with enough time to buy the part, the same availability can be achieved with much less inventory.** The condition is reliability of the signal: removing stock on a weak signal converts savings into downtime.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A mining operation linked vibration-based predictive alerts to MRO purchasing on 87 critical assets. Related spare parts inventory fell 47% and unplanned downtime on those failure modes decreased 34%.

WHAT YOU WOULD USE

> **Buy on condition where the warning is reliable.** Compare warning time with lead time, set triggers, convert stock gradually and close the loop with purchasing.

05 MRO inventory KPIs & cost management

MRO inventory KPIs measure the dual objective of spare parts management: protecting equipment availability while controlling the cost of holding and managing parts.

Inventory turns is a dangerous primary metric for MRO: **high turns can simply mean that critical, slow-moving spares are not being held**. A balanced view weighs availability, total cost — including the estimated cost of downtime — and process quality, so the organization does not optimize one at the expense of the others.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A site praised for high MRO turns suffered high-impact stockouts. After adopting a balanced scorecard it increased stock on 140 high-consequence parts, cut emergency purchases 52% and still achieved a net 11% reduction in MRO inventory value.

WHAT YOU WOULD USE

Measure uptime protection and cost together. Balance availability, total cost and process quality, review jointly with maintenance and watch emergency buys as the early warning.

Inventory Analytics & Technology

Without the right metrics and technology, even good policies and network design degrade. This layer makes inventory visible to operations and to finance.

01 Inventory KPI dashboard & reporting framework

An inventory KPI framework organizes metrics in three levels — operational, tactical and strategic — so each audience sees the measures that match its decisions, from daily replenishment to working capital.

Inventory reporting often either drowns leaders in detail or stays so aggregated that nobody can act. **When inventory is reported only in units and days of supply, it is never managed as working capital.** Connecting service, inventory and cash in one framework lets the monthly conversation move from debating numbers to deciding trade-offs.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A multi-site manufacturer collapsed 47 competing reports into a three-level framework. The monthly conversation shifted from debating numbers to deciding trade-offs, and excess inventory fell $19M while service held.

WHAT YOU WOULD USE

> **Report inventory at three levels, ending in money.** Few metrics per audience, one definition each, financial translation at the top and a cadence that forces decisions.

02 AI & ML for inventory forecasting & optimization

AI and machine learning in inventory improve forecasts for difficult demand patterns — intermittent, promotional, with structural breaks — and help set inventory parameters beyond classical formulas.

Large forecasting competitions have shown that machine-learning methods, particularly gradient-boosted trees, can outperform classical statistical methods on large retail datasets **(Makridakis, Spiliotis and Assimakopoulos, 2022 — International Journal of Forecasting, M5 competition)**. **But better forecasts only reduce inventory if they flow into parameters and decisions**, and the gains depend heavily on data quality.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A consumer goods company applied gradient-boosting models to its top 800 intermittent and promotional SKUs. After feeding the improved forecasts into parameters, inventory on those items fell 14% while service rose 2.1 points.

WHAT YOU WOULD USE

> **Apply ML where classical methods fail, and measure inventory, not accuracy alone.** Clean the data, target hard segments, benchmark on outcomes and wire forecasts into parameters.

03 WMS & inventory accuracy in the warehouse

Inventory accuracy is the degree to which system quantities and locations match physical reality. The WMS is the system of record, but accuracy is ultimately the result of process discipline.

Inaccurate records undermine ATP, replenishment, production scheduling and customer promises; research on retail stores found record inaccuracy to be widespread **(DeHoratius and Raman, 2008 — Management Science)**. **Cycle counting that**

only corrects balances reproduces the same errors indefinitely; eliminating root causes is what raises accuracy durably.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A distribution center shifted from count compliance to root-cause elimination. Location accuracy on A items rose from 93% to 98.9% and pick shortfalls fell 63%; count volume later decreased 35%.

WHAT YOU WOULD USE

> **Count to learn, not just to correct.** Measure accuracy by location, count by segment, investigate every significant variance and close its cause.

04 Inventory digitalization: IoT, RFID & smart shelves

Inventory digitalization uses RFID, IoT sensors and computer vision to create near-real-time, high-accuracy visibility of physical inventory without depending only on manual scans.

Manual processes carry latency and error. **In high-velocity or high-value environments, the cost of inaccuracy can exceed the cost of sensing**; in others it cannot, so selectivity is essential. Retail RFID programs, for example, have been used to raise item-level accuracy where tags are economical.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A high-volume electronics distributor piloted RFID on its top 180 SKUs. Accuracy rose from 94% to 99.4% and pick shortfalls on tagged items dropped 71%; expansion followed only after clear payback.

WHAT YOU WOULD USE

> **Sense where inaccuracy is expensive.** Choose use cases by ROI, match the technology, integrate the signal into actions and scale only after a proven pilot.

05 Inventory financing: working capital & supply chain finance

Inventory financing and supply chain finance treat inventory as a financial asset to be optimized on the balance sheet and in the cash conversion cycle, using metrics such as days inventory outstanding (DIO), cash conversion cycle (CCC) and return on capital employed (ROCE).

Operations usually manages inventory for service while finance manages cash; **when the two are disconnected, decisions that look good operationally can destroy return on capital**. Supply chain finance tools such as reverse factoring can improve cash for both buyer and supplier, but if they only stretch payment terms they move cost and risk to the weaker party.

BEST PRACTICES

EXAMPLE

Illustrative case built from typical industry values — not data from a specific company.

> A manufacturer reduced DIO by 11 days through operational improvements and implemented reverse factoring with its top suppliers, freeing about $38M in cash while structuring the program so suppliers also benefited.

WHAT YOU WOULD USE

Manage inventory in operational and financial terms at once. Report DIO and CCC with service, price the trade-off and use finance tools that help the whole chain, not just one balance sheet.

From the SCRA v2.2 reference architecture · mrsupplychain.ai · © 2026 Ismael López · Mr. Supply Chain® México